TRINAY TELE-OPHTHALMOLOGY NETWORK

AI-ASSISTED RETINAL SCREENING CLINICAL DECISION-SUPPORT REPORT

CONFIDENTIAL MEDICAL DOCUMENT

DATE: 13 Oct 2026

DEPARTMENT OF RETINAL TELE-SCREENING & DIABETIC EYE CARE

1. PATIENT & EXAMINATION DETAILS

Demographics, Examination Parameters & Clinical History

Patient Name / ID: **Aviraj abasaheb Murkute (PX-001)**

Age / Gender: **20 Yrs / Male**

Examined Eye: **Right Eye (OD)**

Examination Date: **13 Oct 2026**

Screening Operator: **Clinic Staff**

Relevant Medical History: Type 2 Diabetes (5 yrs) • Known DR: No

2. SCREENING SUMMARY

Primary AI Staging, Model Score, Image Quality & Review Status

Predicted DR Severity: **Moderate Non-Proliferative DR (Level 2)**

ICDR Classification Standard

Model Output Score: **99%**

Uncalibrated model score; not calibrated probability.

Image Quality & Gradability: **Good (Gradable (Adequate))**

Adequate clarity; no repeat capture indicated.

Clinical Review Status: **Pending Specialist Review**

Queued for tele-ophthalmology verification

3. AI-DETECTED RETINAL FINDINGS

Supported Focal Lesion Findings & Landmark Localizations

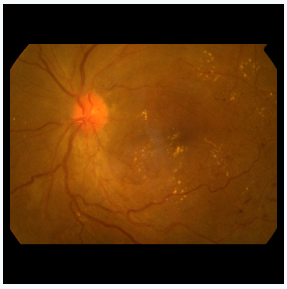
PATHOLOGICAL FINDING	DETECTION STATUS	EXISTING MEASURED RESULT (RAW MODEL UNITS)
Retinal Microaneurysm (MA) (MA)	Not evaluated	Unavailable
Retinal Hemorrhage (HE) (HE)	Not evaluated	Unavailable
Hard Exudate (EX) (EX)	Not evaluated	Unavailable
Soft Exudate / Cotton-Wool Spot (SE) (SE)	Not evaluated	Unavailable

RETINAL ANATOMY LOCALIZATION:

Optic Disc: Localized • Fovea: Identified • Macula: Estimated ROI (r=56px) • Vessels: Segmented

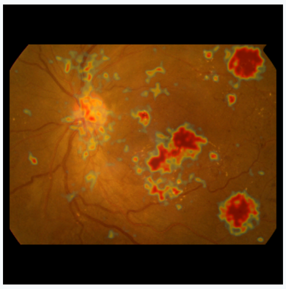
4. VISUAL EVIDENCE

4 Canonical Model Inference Outputs (Proportionally Scaled Without Cropping)



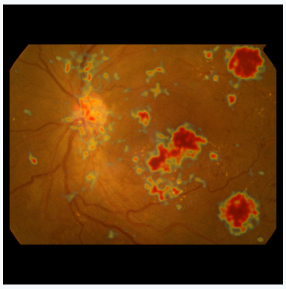
1. Original Fundus

True-colour reference



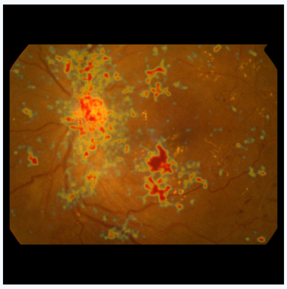
2. Combined AI Overlay

Lesions & landmarks



3. Grad-CAM++

Neural activation map



4. Retinal Anatomy

Disc, fovea & vessels

5. AI SCREENING INTERPRETATION

Status: Clinician Review Required

Lesion Segmentation Findings Unavailable • Predicted DR Category: Moderate Non-Proliferative DR (Level 2)

Whole-image neural classifier staging indicates Moderate Non-Proliferative DR (Level 2). Focal lesion segmentation was not evaluated or recorded for this image.

Model Analysis: Direct concordance comparison between classification and lesion burden cannot be performed without segmentation outputs. Clinician review required.

All reported metrics reflect automated computer-vision model inferences and 2D segmentation masks. They do not constitute confirmed clinical diagnoses or replace in-person biomicroscopy.

6. RECOMMENDED CLINICAL ACTION

Prompt Specialist Review

Comprehensive Ophthalmology Referral (Prompt Evaluation)

Refer to a vitreo-retinal specialist for comprehensive dilated fundus examination, slit-lamp biomicroscopy, and macular optical coherence tomography (OCT).

Recommended Follow-Up Interval: Within 2 to 4 weeks (Prompt Specialist Review)

Guideline Basis: International Council of Ophthalmology (ICO) Guidelines for Severe / Proliferative DR • AI recommendation subject to clinician confirmation

7. OPHTHALMOLOGIST REVIEW

Status: Pending Specialist Review

Reviewing Specialist:	**Assigned Review Queue: Tele-Ophthalmology Network**	Review Date:	**13 Oct 2026**
Clinical Assessment:	**Pending specialist clinical counter-verification**		
Assessment Notes:	Queued for ophthalmologist counter-verification and review.		

REVIEWING OPHTHALMOLOGIST:	MEDICAL REGISTRATION / LICENSE:	CLINICAL SIGNATURE & DATE:
Dr. ____________________________	**MCI / Reg:** _____________________	**Date:** _________________________

8. DISCLAIMER

TRINAY is an artificial intelligence decision-support tool intended for tele-retinal screening and primary triage. It does not provide an autonomous medical diagnosis. All automated classifications, lesion segmentations, and landmark localizations must be clinically correlated and verified by a licensed ophthalmologist through direct clinical examination (slit-lamp biomicroscopy / OCT).

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